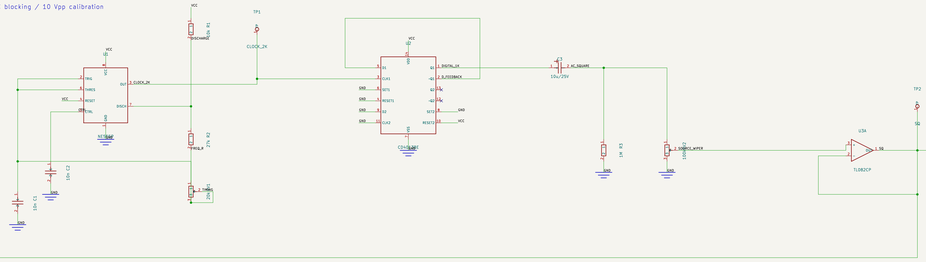
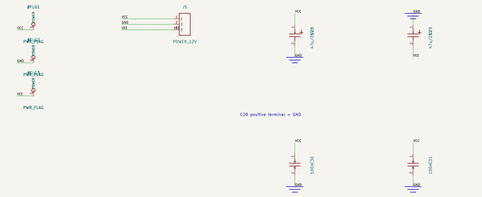
NE555 + TL082 FOURIER WAVEFORM SYNTHESIS — Rev B wired schematic

Regulated +12 V / GND / −12 V. Signal wires are drawn explicitly. All labels are single-sheet local nets.

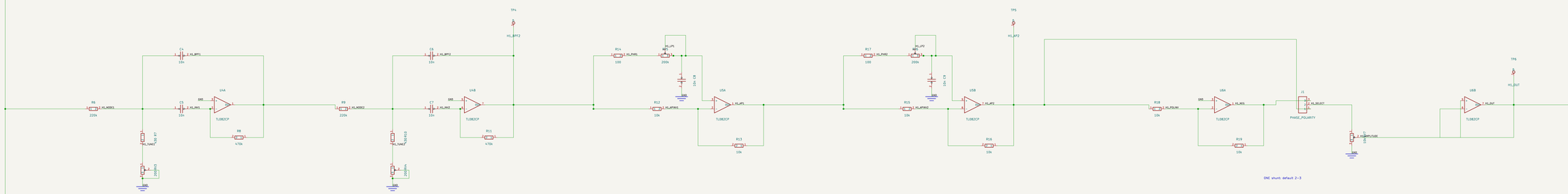
SOURCE: 2 kHz clock / divide by two / DC blocking / 10 Vpp calibration



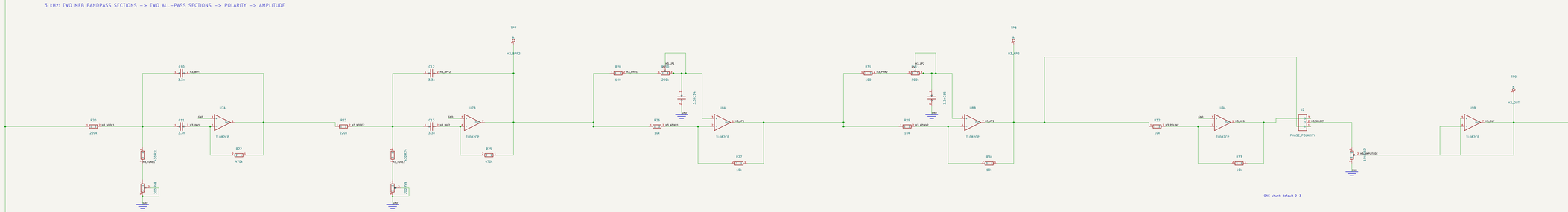
REGULATED +/−12 V INPUT



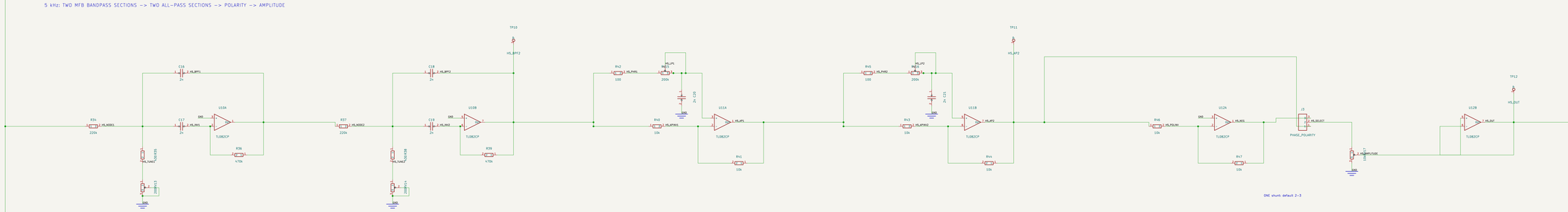
1 kHz: TWO MFB BANDPASS SECTIONS → TWO ALL-PASS SECTIONS → POLARITY → AMPLITUDE



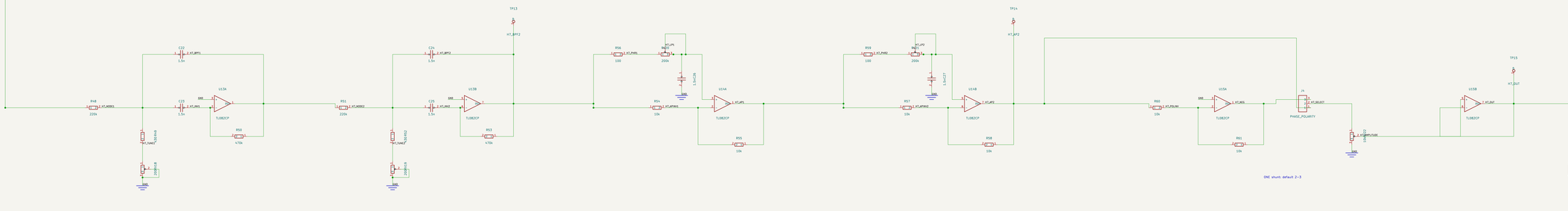
3 kHz: TWO MFB BANDPASS SECTIONS → TWO ALL-PASS SECTIONS → POLARITY → AMPLITUDE



5 kHz: TWO MFB BANDPASS SECTIONS → TWO ALL-PASS SECTIONS → POLARITY → AMPLITUDE



7 kHz: TWO MFB BANDPASS SECTIONS → TWO ALL-PASS SECTIONS → POLARITY → AMPLITUDE



SUM AND OUTPUT

ANALOG POWER AND LOCAL BYPASS: 100 nF at each supply pin

Sheet: /
File: waveform.kicad_sch

Title: Wired Fourier waveform synthesis

Size: A3 | Date: 2026-11-30 | Rev: B

KiCad 9.0.4, 9.0.2 | Id: 1/1